## V4.5 Comprehensive Statistical Summary

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CAR V4.5 CENTERED MODEL AUDIT | idata_car_v4.5.nc
================================================================
WAIC (Deviance): 4221.58 | LOO-CV: 4287.24
Chains: 4 | Samples: 3000 | Divergences: 0
Max R-hat: 1.790

V4.5 SPECIFICATION:
- Effect Coding: Men (+0.5) / Women (-0.5)
- Centered Interaction: Smoke * (Gender - 0.5)
- Variance Anchor: InverseGamma(3.0, 1.0)

PUBLIC HEALTH INTERPRETATION:
- Population Smoking Effect: +1.92% risk per 5.3% increase
- Gender Modifier (Men): -0.30% additional risk adjustment
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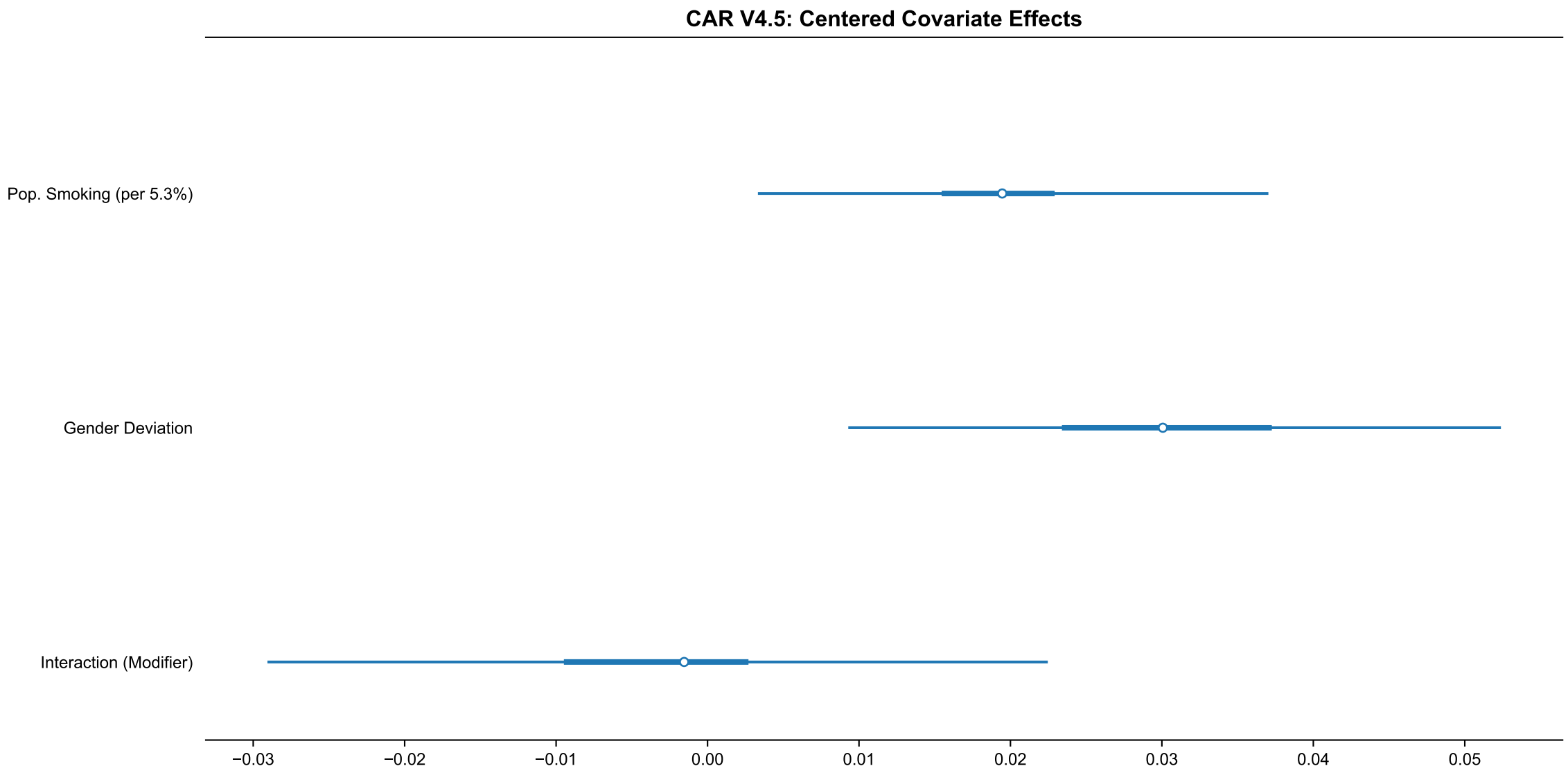
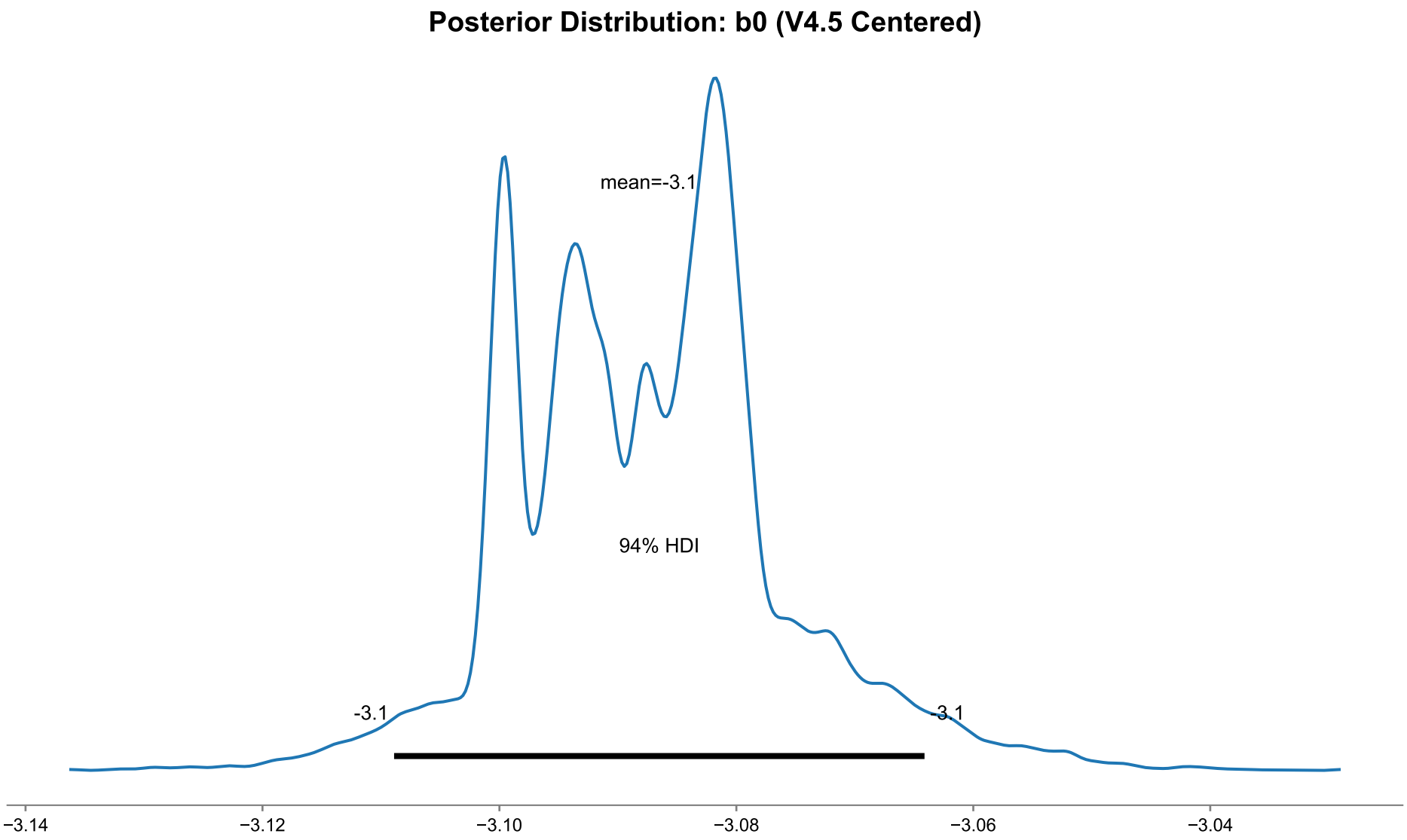
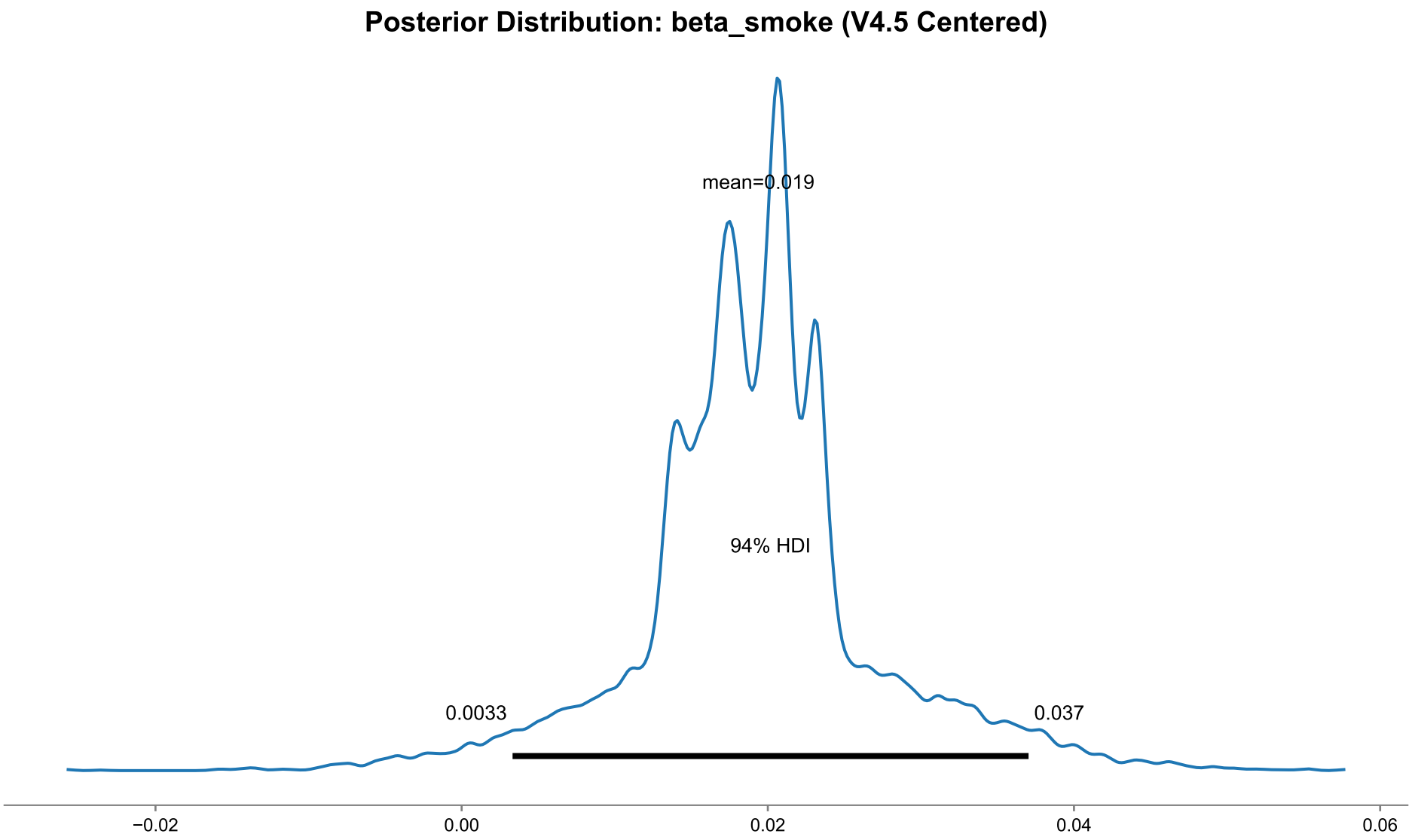
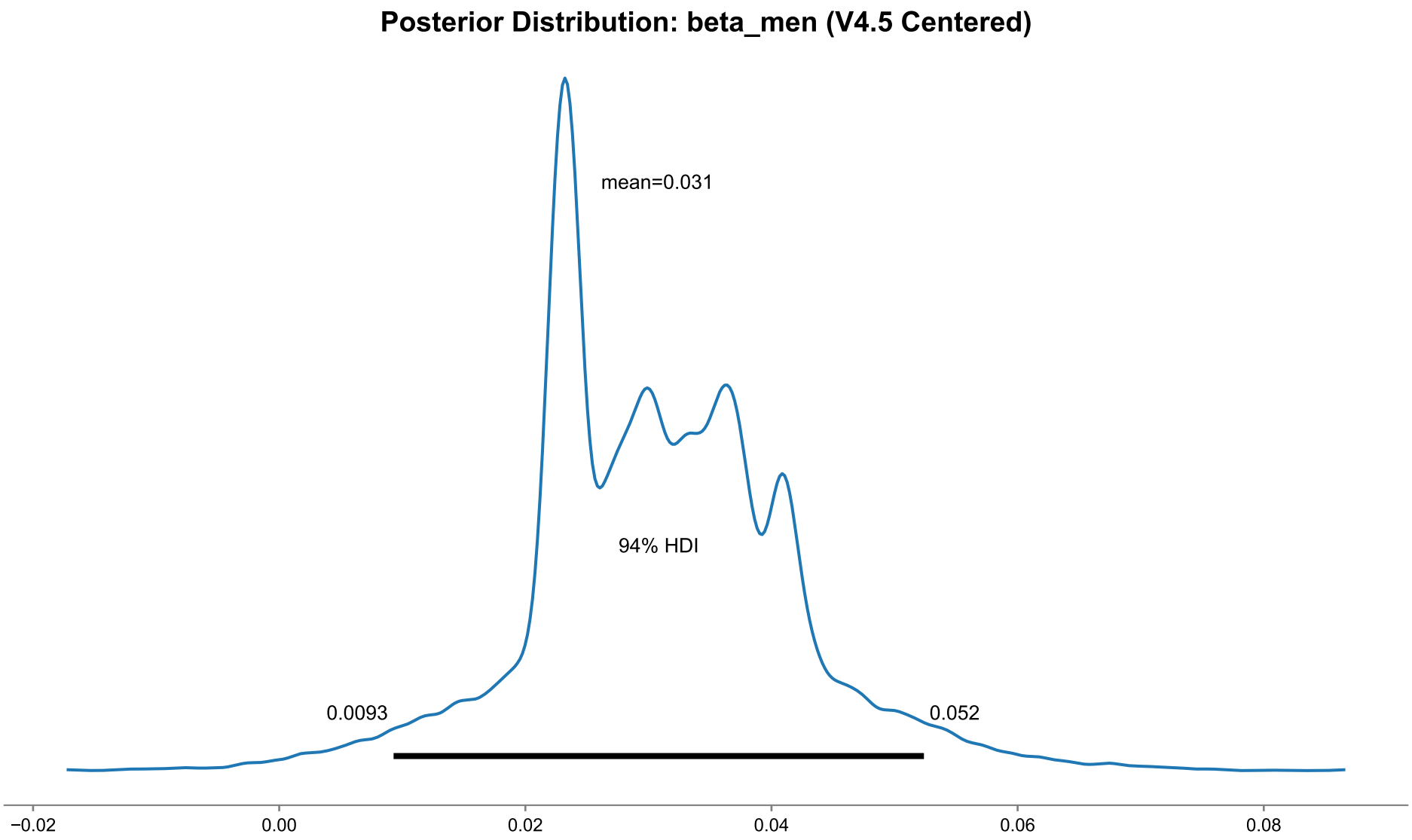
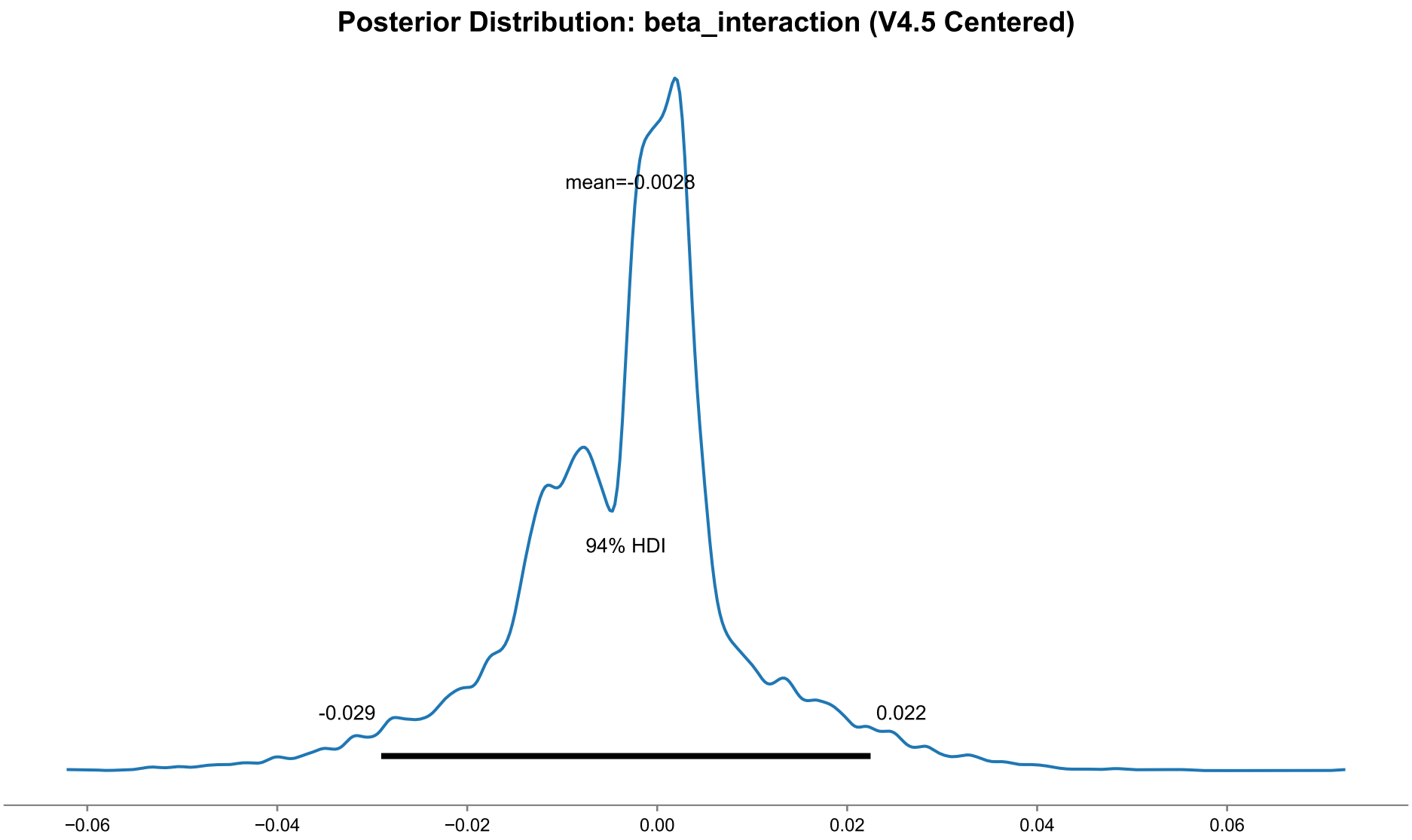
| | mean | sd | hdi_3% | hdi_97% | mcse_mean | mcse_sd | ess_bulk | ess_tail | r_hat |
|---|---|---|---|---|---|---|---|---|---|
| b0 | -3.087 | 0.012 | -3.109 | -3.064 | 0.003 | 0.002 | 16.0 | 105.0 | 1.20 |
| beta_smoke | 0.019 | 0.008 | 0.003 | 0.037 | 0.001 | 0.002 | 35.0 | 102.0 | 1.25 |
| beta_men | 0.031 | 0.011 | 0.009 | 0.052 | 0.002 | 0.002 | 22.0 | 102.0 | 1.18 |
| beta_interaction | -0.003 | 0.012 | -0.029 | 0.022 | 0.002 | 0.003 | 35.0 | 105.0 | 1.38 |
| sigma | 0.258 | 0.026 | 0.217 | 0.297 | 0.010 | 0.004 | 8.0 | 110.0 | 1.51 |
| rho | 0.098 | 0.124 | 0.000 | 0.358 | 0.027 | 0.020 | 37.0 | 124.0 | 1.79 |
| alpha | 0.514 | 0.213 | 0.128 | 0.998 | 0.028 | 0.045 | 40.0 | 97.0 | 1.37 |

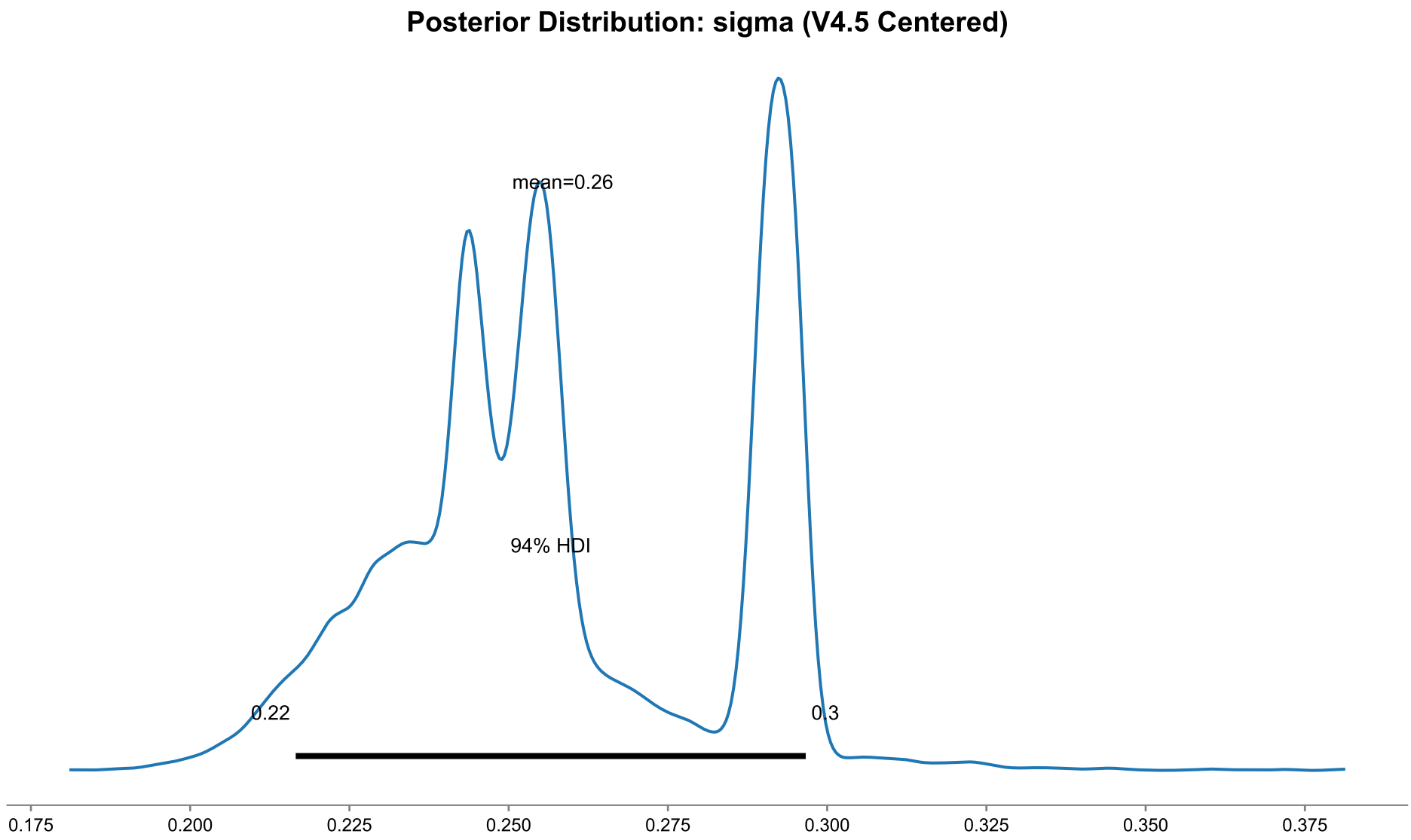

CAR V4.5: Centered Covariate Effects
Pop. Smoking (per 5.3%)
Gender Deviation
Interaction (Modifier)
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
0.05

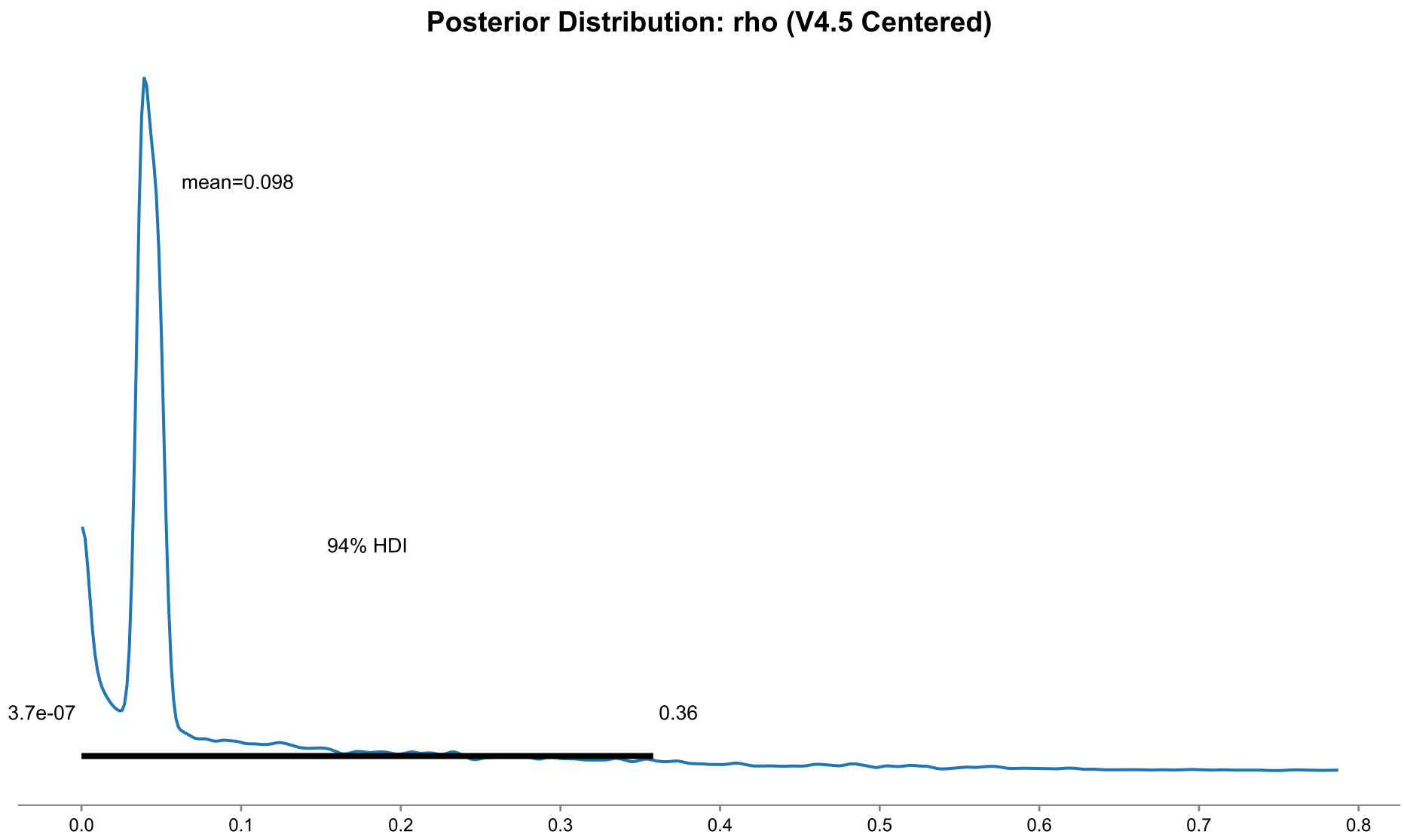

Posterior Distribution: b0 (V4.5 Centered)
mean=-3.1
94% HDI
-3.1
-3.1
−3.14
−3.12
−3.10
−3.08
−3.06
−3.04

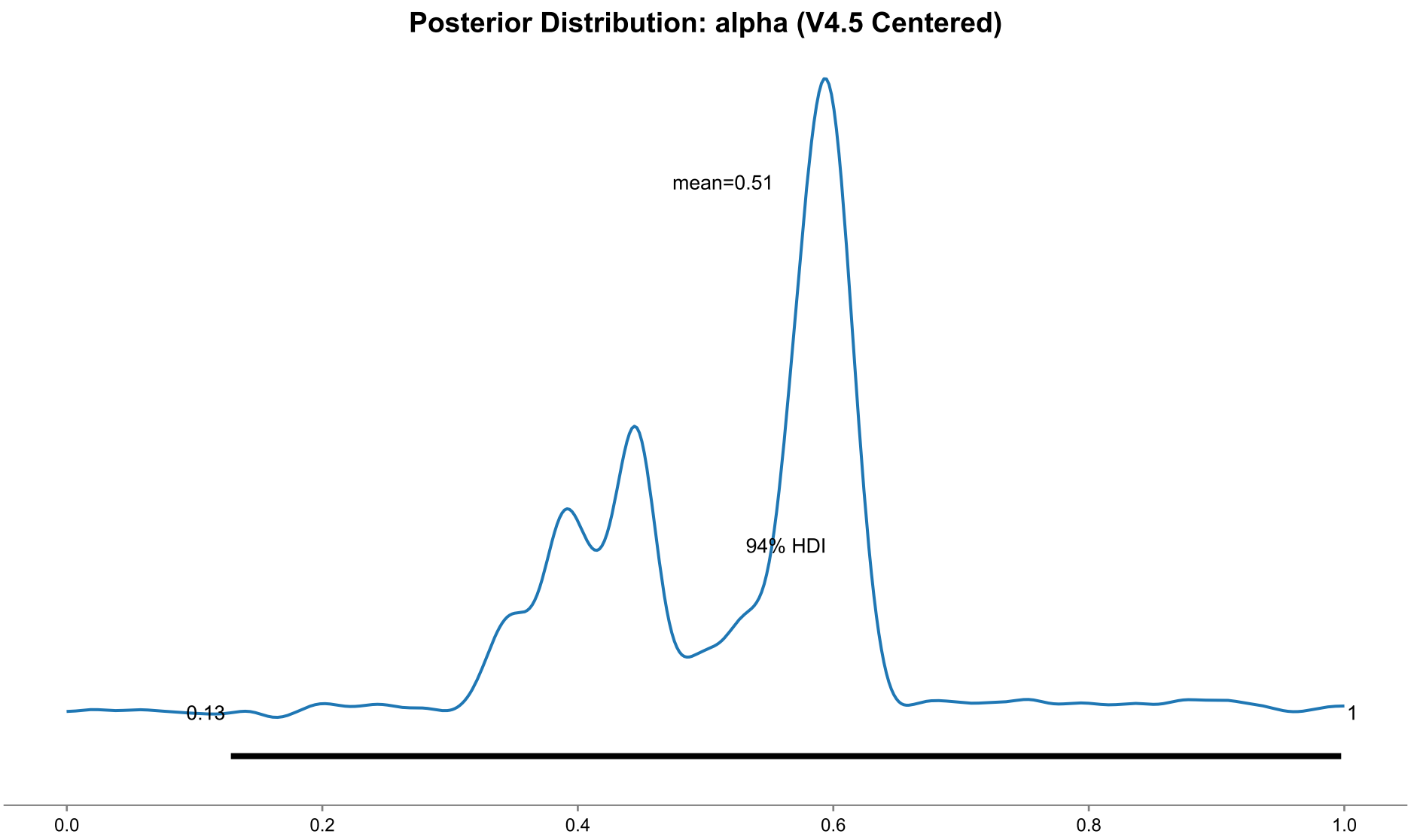
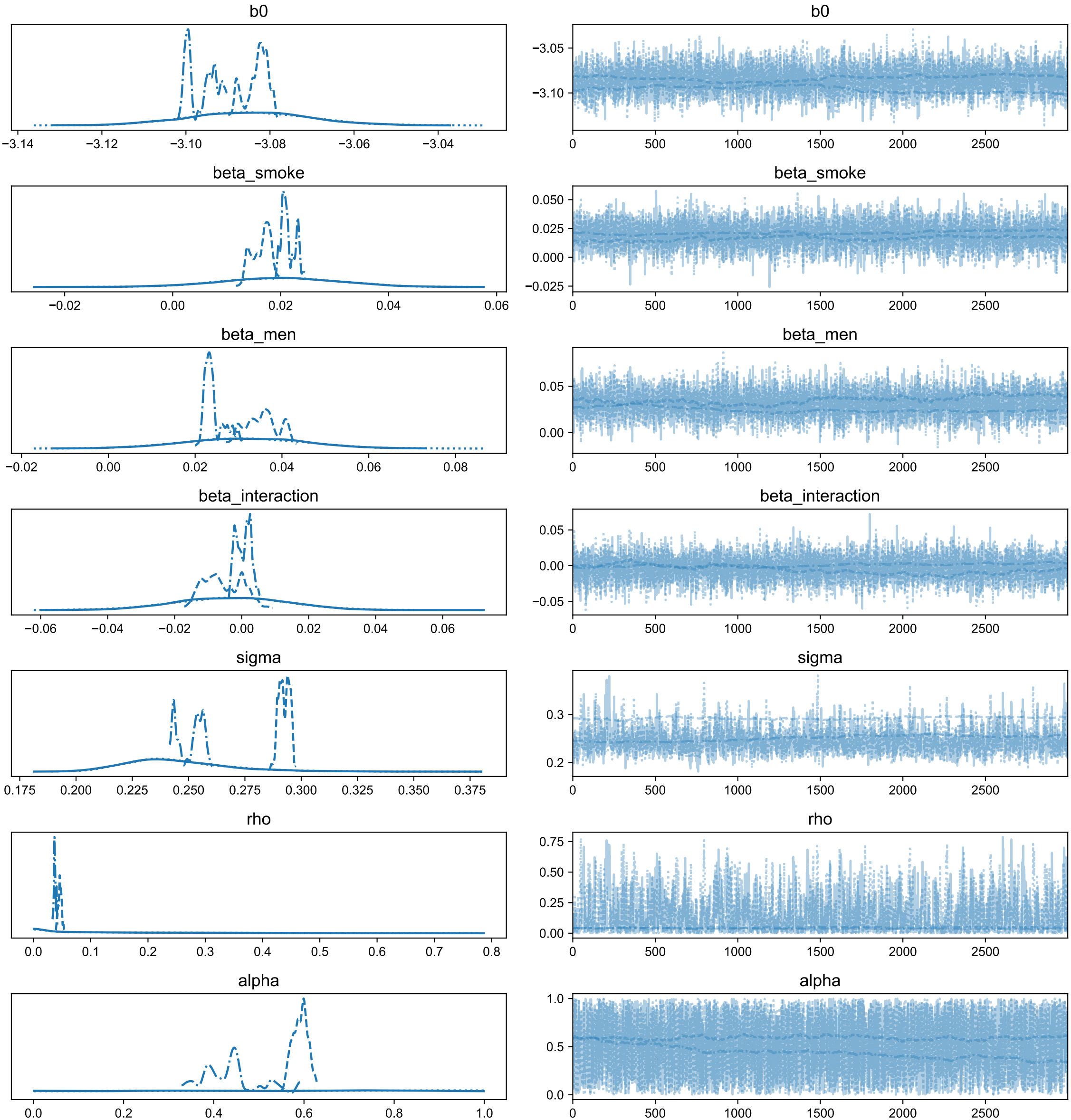

Posterior Distribution: beta_smoke (V4.5 Centered)
mean=0.019
94% HDI
0.0033
0.037
−0.02
0.00
0.02
0.04
0.06


Posterior Distribution: beta_men (V4.5 Centered)
mean=0.031
94% HDI
0.0093
0.052
−0.02
0.00
0.02
0.04
0.06
0.08


Posterior Distribution: beta_interaction (V4.5 Centered)
mean=-0.0028
94% HDI
-0.029
0.022
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06


Posterior Distribution: sigma (V4.5 Centered)
mean=0.26
94% HDI
0.22
0.3
0.175
0.200
0.225
0.250
0.275
0.300
0.325
0.350
0.375


Posterior Distribution: rho (V4.5 Centered)
mean=0.098
94% HDI
3.7e-07
0.36
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8


Posterior Distribution: alpha (V4.5 Centered)
mean=0.51
94% HDI
0.13
1
0.0
0.2
0.4
0.6
0.8
1.0

# V4.5 MCMC Chain Traces